Chapter 3 : Signed Measures and Differentiation

Arkaraj Mukherjee

April 12, 2026

Contents

Chapter 3 : Signed Measures and Differentiation **3**
Problem 3.1 3
Problem 3.2 3
Problem 3.3 3
Problem 3.4 4
Problem 3.5 4
Problem 3.6 4
Problem 3.7 4
Problem 3.8 4
Problem 3.9 5
Problem 3.10 5
Problem 3.11 5
Problem 3.12 6
Problem 3.13 6
Problem 3.14 6
Problem 3.15 7
Problem 3.16 8
Problem 3.17 8
Problem 3.18 8
Problem 3.19 9
Problem 3.20 9
Problem 3.21 9
Problem 3.22 10
Problem 3.23 10
Problem 3.24 10
Problem 3.25 10
Problem 3.26 10
Problem 3.27 10
Problem 3.28 10
Problem 3.29 11
Problem 3.30 11
Problem 3.31 11
Problem 3.32 11
Problem 3.33 11
Problem 3.34 11
Problem 3.35 12
Problem 3.36 12
Problem 3.37 12
Problem 3.38 12
Problem 3.39 12
Problem 3.40 12
Problem 3.41 13
Problem 3.42 13

Chapter 3 : Signed Measures and Differentiation

Problem 3.1. Prove Proposition 3.1.

Solution. Wlog suppose that ν omits the value $-\infty$, then if E_n is a sequence increasing sets we have that,

$$\nu\left(\bigcup_{n=1}^{\infty} E_n\right) = \nu(E_1) + \sum_{n\geq 2} \nu(E_n \setminus E_{n-1}) = \lim_{m\to\infty} \nu(E_1) + \sum_{n\leq m} \nu(E_n \setminus E_{n-1}) = \lim_{m\to\infty} \nu(E_m)$$

And if it was decreasing with $\nu(E_1)$ finite then, $E_1 \setminus E_n$ is increasing and we have

$$\nu(E_1) - \nu\Big(\bigcap E_n\Big) = \nu\Big(E_1 \setminus \bigcap E_n\Big) = \nu\Big(\bigcup (E_1 \setminus E_n)\Big) = \lim_{m\to\infty} \nu(E_1 \setminus E_m) = \nu(E_1) - \lim_{m\to\infty} \nu(E_m)$$

after which we can cancel out the finite term of $\nu(E_1)$. $\square$

Problem 3.2. If ν is a signed measure, E is ν-null iff $|\nu|(E) = 0$. Also, if ν and μ are signed measures, $\nu \perp \mu$ iff $|\nu| \perp \mu$ iff $\nu^+ \perp \mu$ and $\nu^- \perp \mu$.

Solution. If $|\nu|(E) = 0$ then $\nu^+(E) = \nu^-(E) = 0$ and as these are positive measures, by monotonicity see that $\nu^+(F) = \nu^-(F) = 0$ for any measurable $F \subseteq E$ making E ν-null. Now, if E was ν-null then $|\nu|(E) > 0$ would imply that $\nu^+(E)$ or $\nu^-(E)$ is strictly greater than 0 and if we assume the first to be as described then $\nu(E \cap N) = \nu^+(E \cap N) > 0$ which contradicts nullity of E w.r.t ν as $E \cap N \subseteq E$. The other case leads to a contradiction similarly, thus we must have $|\nu|(E) = 0$.

As a measurable set E is $|\nu|$-null iff $|\nu|(E) = 0$ stemming from positivity we see that $\nu \perp \mu$ iff $|\nu| \perp \mu$ by the previous claim. We also see that $|\nu| \perp \mu$ iff $\nu^+, \nu^- \perp \mu$ as $|\nu|(E) = 0$ iff $\nu^-(E), \nu^+(E) = 0$, which proves the equivalence. $\square$

Problem 3.3. Let ν be a signed measure on $(X, \mathcal{M})$.

a. $L^1(\nu) = L^1(|\nu|)$.

b. If $f \in L^1(\nu)$, $|\int f \, d\nu| \leq \int |f| d|\nu|$.

c. If $E \in \mathcal{M}$, $|\nu|(E) = \sup\{|\int_E f \, d\nu| : |f| \leq 1\}$.

Solution.

a. As $L^1(\nu) := L^1(\nu^+) \cap L^1(\nu^-)$ we find that, $f \in L^1(\nu)$ implies $\int |f|, \mathrm{d}\nu^+, \int |f| \, \mathrm{d}\nu^- < \infty$ and thus, $\int |f| \, \mathrm{d}|\nu| = \int |f| \, \mathrm{d}\nu^+ + \int |f| \, \mathrm{d}\nu^- < \infty$ where we have used the following lemma: If μ_1, μ_2 are positive measures on the same sigma algebra then so is their sum and if $f \in L^+$ then $\int f \, \mathrm{d}(\mu_1 + \mu_2) = \int f \, \mathrm{d}\mu_1 + \int f \, \mathrm{d}\mu_2$. For a proof, one can look at simple functions first where its trivial and then routinely extend to L^+ by MCT and approximations using simple functions. Now, if we had $f \in L^1(|\nu|)$ then as $\int f \, \mathrm{d}|\nu| = \int |f| \, \mathrm{d}\nu^+ + \int |f| \, \mathrm{d}\nu^- < \infty$, both of these terms are individually finite which puts f in $L^1(\nu^+)$ and $L^1(\nu^-)$ proving the claim.

b. This follows from the inequality $\int |f| \, \mathrm{d}\nu = \int |f| \, \mathrm{d}\nu^+ - \int |f| \, \mathrm{d}\nu^- \leq \int |f| \, \mathrm{d}\nu^+ + \int |f| \, \mathrm{d}\nu^- = \int |f| \, \mathrm{d}|\nu|$.

c. Consider the hahn decomposition $P \cup N$ is usual notation leading upto the jordan decomposition, then letting $f = \chi_P - \chi_N$ we see that $\int f \, \mathrm{d}\nu = \nu(P \cap E) - \nu(N \cap E) = \nu^+(E) + \nu^-(E) = |\nu|(E)$ so we have one direction of the inequality ($\leq$) and for the other direction, by part (b) we can say that $|f| \leq 1$ implies $|\int f \, \mathrm{d}\nu| \leq \int 1 \, \mathrm{d}|\nu| = |\nu|(E)$. so we are done.

□

Problem 3.4. If ν is a signed measure and λ, μ are positive measures such that $\nu = \lambda - \mu$, then $\lambda \geq \nu^+$ and $\mu \geq \nu^-$.

Solution. Consider the hahn decomposition $P \cup N$ in usual notation, leading upto the jordan decomposition. Then we have $\nu^+(E) = \nu(E \cap P) = \lambda(E \cap P) - \mu(E \cap P) \leq \lambda(E \cap P) \leq \lambda(E)$ and the other inequality follows similarly. □

Problem 3.5. If ν_1, ν_2 are signed measures that both omit the value $+\infty$ or $-\infty$, then $|\nu_1 + \nu_2| \leq |\nu_1| + |\nu_2|$. (Use Exercise 4.)

Solution. It can be easily shown that in this case $\nu_1 + \nu_2$ is also a signed measure. As all ν_1, ν_2 are finite, we can rearrange to get $\nu_1 + \nu_2 = \nu_1^+ - \nu_1^- + \nu_2^+ - \nu_2^- = (\nu_1^+ + \nu_2^+) - (\nu_1^- + \nu_2^-)$ which, by the previous problem implies $(\nu_1 + \nu_2)^+ \leq \nu_1^+ + \nu_2^+$ and $(\nu_1 + \nu_2)^- \leq \nu_1^- + \nu_2^-$ implying $|\nu_1 + \nu_2| \leq (\nu_1^+ + \nu_2^+) + (\nu_1^- + \nu_2^-) = (\nu_1^+ + \nu_1^-) + (\nu_2^+ + \nu_2^-) \leq |\nu_1| + |\nu_2|$. □

Problem 3.6. Suppose $\nu(E) = \int f \, d\mu$ where μ is a positive measure and f is an extended μ-integrable function. Describe the Hahn decompositions of ν and the positive, negative, and total variations of ν in terms of f and μ.

Solution. Let $P = \{f > 0\}$ and $N = \{f \leq 0\}$, then $P \cup N$ is clearly a hahn decomposition and we see that $\nu_+(E) = \int_{E \cap P} f \, \mathrm{d}\mu = \int_E f^+ \, \mathrm{d}\mu$ and similarly $\nu_-(E) = \int_E f^- \, \mathrm{d}\mu$ and thus, $|\nu|(E) = \int_E |f| \, \mathrm{d}\mu$. □

Problem 3.7. Suppose that ν is a signed measure on $(X, \mathcal{M})$ and $E \in \mathcal{M}$.

a. $\nu^+(E) = \sup\{\nu(F) : F \in \mathcal{M}, F \subset E\}$ and $\nu^-(E) = -\inf\{\nu(F) : F \in \mathcal{M}, F \subset E\}$.

b. $|\nu|(E) = \sup\{\sum_1^n |\nu(E_j)| : n \in \mathbb{N}, E_1, ..., E_n$ are disjoint, and $\bigcup_1^n E_j = E\}$.

Solution.

a. $F \subseteq E$ implies that $\nu(F) \leq \nu^+(F) \leq \nu^+(E)$ and $\nu(E \cap P) = \nu^+(E)$ with $E \cap P \subseteq E$ proving one part, the other follows similarly.

b. Considering a hahn decomposition $P \cup N$ leading to a jordan decomposition, we see that $|\nu|(E) = \nu^+(E) + \nu^-(E) = |\nu(E \cap P)| + |\nu(E \cap N)|$ as $\nu^+(E) = \nu(E \cap P) \geq 0$ and $\nu^-(E) = -\nu(E \cap N) \geq 0$ proving one direction of the equality. For the other, it can be easily proved that $|\nu(E)| \leq |\nu|(E)$ and thus, $\bigcup_1^n E_j = E$ (disjoint union) implies $\sum_1^n |\nu(E_j)| \leq \sum_1^n |\nu|(E_j) = |\nu|(E)$ so we are done.

□

Problem 3.8. $\nu \ll \mu$ iff $|\nu| \ll \mu$ iff $\nu^+ \ll \mu$ and $\nu^- \ll \mu$.

Solution. If $\nu \ll \mu$ then $\mu(E) = 0 \implies \nu(E) = 0$ and as μ is taken positive we must have that $\mu(E \cap P), \mu(E \cap N) = 0$ where $P \cup N$ is the hahn decomposition for ν, this implies that $\nu(E \cap P), \nu(E \cap N) = 0$ so $\nu^{\pm}(E) = 0$ and we have that $|\nu|(E) = 0$ as well proving $|\nu| \ll \mu$. As $|\nu|(E) \geq |\nu(E)|$ we have that $|\nu| \ll \mu$ implies $\nu \ll \mu$ and hence $\nu \ll \mu$ iff $|\nu| \ll \mu$. If $\nu^+\nu^- \ll \mu$ then $\mu(E) = 0$ clearly implies $|\nu|(E) = 0$ and if we have $|\nu| \ll \mu$ then $\mu(E) = 0$ implies $\nu^+(E) + \nu^-(E) = 0$ and thus $\nu^+(E) = \nu^-(E) = 0$ proving $\nu^+, \nu^- \ll \mu$ and thus all these conditions are equivalent. □

Problem 3.9. Suppose $\{\nu_j\}$ is a sequence of positive measures. If $\nu_j \perp \mu$ for all j, then $\sum_1^\infty \nu_j \perp \mu$; and if $\nu_j \ll \mu$ for all j, then $\sum_1^\infty \nu_j \ll \mu$.

Solution. Say E_j is ν_j-null and E_j^c is μ-null for all j, then we see that, let $E = \bigcap E_j$, then we clearly see that E is $\sum_1^\infty \nu_j$-null and we claim that, $E^c = \bigcup E_j^c$ is μ-null. For any measurable $F \subseteq E^c$, we can write $F = \bigcup_{j=1}^\infty (F \cap (E_j^c \setminus \cup_{i \neq j}^\infty E_i^c))$ is a disjoint union and

$$\mu(F) = \sum_{j=1}^\infty \mu(F \cap (E_j^c \setminus \cup_{i \neq j}^\infty E_i^c)) = 0$$

as $(F \cap (E_j^c \setminus \cup_{i \neq j}^\infty E_i^c)) \subseteq E_j^c$. □

Problem 3.10. Theorem 3.5 may fail when ν is not finite. (Consider $d\nu(x) = dx/x$ and $d\mu(x) = dx$ on $(0, 1)$, or $\nu =$ counting measure and $\mu(E) = \sum_{n \in E} 2^{-n}$ on $\mathbb{N}$.)

Solution. We choose the second counterexample. Clearly we have, $\nu \ll \mu$ but for any $\delta > 0$ we see that there exists a E_δ with $\mu(E_\delta) < \delta$ but $\nu(E_\delta) \geq 1$ which just reads "E is nonempty" and as the sum $\sum_{n \in \mathbb{N}} 2^{-n}$ converges absolutely we can find such a E_δ always. □

Problem 3.11. Let μ be a positive measure. A collection of functions $\{f_\alpha\}_{\alpha \in A} \subset L^1(\mu)$ is called uniformly integrable if for every $\epsilon > 0$ there exists $\delta > 0$ such that $|\int_E f_\alpha d\mu| < \epsilon$ for all $\alpha \in A$ whenever $\mu(E) < \delta$.

a. Any finite subset of $L^1(\mu)$ is uniformly integrable.

b. If $\{f_n\}$ is a sequence in $L^1(\mu)$ that converges in the L^1 metric to $f \in L^1(\mu)$, then $\{f_n\}$ is uniformly integrable.

Solution.

a. Consider $A = \{1, \ldots, n\}$; for any $\epsilon > 0$, as $f_j d\mu \ll \mu$ where $f_j d\mu$ are finite meausres, we can find $\delta_j > 0$ such that $|\int_E f_j d\mu| < \epsilon$ for all $j \in A$ whenever $\mu(E) < \delta_j$ so taking $\delta = \min_j \delta_j$ suffices.

b. As f_n converges in L^1, it is cauchy in this metric too so for any $\epsilon > 0$ we can find N such that $m, n \geq N \implies |\int_E f_m - \int_E f_n| \leq \int |f_m - f_n| < \epsilon$ and by the previous problem, as $\{f_1, \ldots, f_N\}$ are uniformly integrable, we can find $\delta > 0$ such that, $|\int_E f_j| < \epsilon < 2\epsilon$ for $1 \leq j \leq N$ whenever $\mu(E) < \delta$ and for $n > N$,

$$\left|\int_E f_n d\mu\right| \leq \left|\int_E f_N d\mu\right| + \left|\int_E f_n - f_N d\mu\right| < \epsilon + \epsilon = 2\epsilon$$

so we are done.

□

Problem 3.12. For $j = 1, 2$, let μ_j, ν_j be σ-finite measures on $(X_j, \mathcal{M}_j)$ such that $\nu_j \ll \mu_j$. Then $\nu_1 \times \nu_2 \ll \mu_1 \times \mu_2$ and

$$\frac{d(\nu_1 \times \nu_2)}{d(\mu_1 \times \mu_2)}(x_1, x_2) = \frac{d\nu_1}{d\mu_1}(x_1)\frac{d\nu_2}{d\mu_2}(x_2).$$

Solution. Suppose $\mu_1 \times \mu_2(E) = 0$ then we see that $0 = \int \mu_1(E_x)d\mu_2(x)$ implying $\mu_1(E_x) = 0$ (this in turn implies $\nu_1(E_x) = 0$ here) for μ_2-a.e. x so suppose its true for all x outside N with $\mu_2(N) = 0$ which implies $\nu_2(N) = 0$ so we have that $\nu_1 \times \nu_2(E) = \int \nu_1(E_x)d\nu_2(x) = 0$ as $\nu_1(E_x) = 0$ for ν_2-a.e. x thus proving $\nu_1 \times \nu_2 \ll \mu_1 \times \mu_2$. CLearly these product measures are sigma finite too, so we can find a radon nikodyn derivative f i.e. $d(\nu_1 \times \nu_2) = fd(\mu_1 \times \mu_2)$ and also f_1, f_2 such that $f_j d\mu_j = \nu_j$ for $j = 1, 2$ and by fubini's we see that,

$$\int \chi_E(x, y)f_1(x)f_2(y)d(\mu_1 \times \mu_2(x, y)) = \int\int f_1(x)\chi_{E_x}(y)f_2(y)d\mu_2(y)d\mu_1(x)$$

$$= \int \nu_2(E_x)f_1(x)d\mu_1(x) = \int \nu_2(E_x)d\nu_1(x) = \nu_1 \times \nu_2(E) = \int \chi_E f(x, y)d(\mu_1 \times \mu_2(x, y))$$

which proves that $f(x, y) = f_1(x)f_2(y)$ for $\mu_1 \times \mu_2$ a.e. (x, y) so we have the equality treating them as equivalence classes. □

Problem 3.13. Let $X = [0, 1]$, $\mathcal{M} = \mathcal{B}_{[0,1]}$, $m =$ Lebesgue measure, and $\mu =$ counting measure on $\mathcal{M}$.

a. $m \ll \mu$ but $dm \neq f\ d\mu$ for any f.

b. μ has no Lebesgue decomposition with respect to m.

Solution.

a. As $\mu(E) = 0 \implies E = \emptyset$, $m \ll \nu$ holds. If we had $dm = fd\mu$ then we would have $m(\{x\}) = f(x) = 0$ for all x which is absurd as it implies $dm = 0$.

b. If we had $\mu = \rho + \lambda$ where $\lambda \perp m$ and $\rho \ll m$ then if $[0, 1] = M \cup L$ where L, M are m, λ-null repsectively then as $m(L) = 0$ we have that $m(M) = 1$ and thus, $\infty = \mu(M) = \rho(M) + \mu(M) = 0 + 1 = 1$ which is not possible.

□

Problem 3.14. If ν is an arbitrary signed measure and μ is a σ-finite measure on $(X, \mathcal{M})$ such that $\nu \ll \mu$, there exists an extended μ-integrable function $f : X \to [-\infty, \infty]$ such that $d\nu = f\ d\mu$. Hints:

a. It suffices to assume μ is finite and ν is positive.

b. With these assumptions there exists $E \in \mathcal{M}$ that is σ-finite for ν and $\mu(E) \geq \mu(F)$ for all sets F that are σ-finite for ν.

c. The radon-nikodym theorem applies on E. If $F \cap E = 0$, then either $\nu(F) = \mu(F) = 0$ or $\mu(F) > 0$ and $|\nu(F)| = \infty$.

Solution. If this is true for positive ν and finite μ, then for sigma finite μ we can write $X = \bigcup_1^\infty E_n$ with E_n being μ-finite and mutually disjoint we have $d\nu_n = f_n d\mu_n$ (we safely set $f_n = 0$ outside E_n) where $\mu_n(E) := \mu(E \cap E_n)$ and $\nu_n(E) = \nu(E \cap E_n)$ as μ_n is finite and clearly $\nu_n \ll \mu_n$ still holds. Then, with $f = \sum_1^\infty f_n$ we would have $d\nu = f d\mu$. If ν was signed on top of that, then using Problem 3.8 which states $\nu \ll \mu \implies \nu^+, \nu^- \ll \mu$ we can find f^+, f^- with $d\nu^- = f^- d\mu, d\nu^+ = f^+ d\mu$ we have $d\nu = d\nu^+ - d\nu^- = (f^+ - f^-)d\mu$ and $f^+ - f^-$ is clearly extended integrable as ν, by definition can only assume atmost one of the values $+\infty, -\infty$. We now prove it for this case. Consider the set $\{\mu(H) : H \in \mathcal{M}, H \text{ is } \nu\, \sigma\text{-finite}\}$, let s be its supremum, then $s < \infty$ as μ is finite and we can find a ν σ-finite sequence of sets F_n such that $\mu(F_n) \longrightarrow s$, then we can clearly see that $\mu(\cup F_n) = s$ as $\cup F_n$ is clearly still ν σ-finite and we can show both sides of the equality easily. Setting $E = \cup F_n$ gives us a set as described in hint (b). The radon nikodym theorem applies on E clearly. If $F \cap E = \emptyset$ then if we have $\nu(F) > 0$ we must have $\mu(F) > 0$ as $\nu \ll \mu$which forces $\nu(F) = \infty$ as otherwise the existence of $E \cup F$ contradicts the maximality of E and if we have $\mu(F) > 0$ then again we similarly have $\nu(F) = \infty$ such as to not contradict the maximality of E so we have proved the claim in hint (c). There are now two cases, first one being whem $\nu(E^c) = \mu(E^c) = 0$, in this case we can apply radon-nikodym to get a derivative f on E and extend it to the rest of the set by setting it to be zero outside this, then we see that,

$$\nu(K) = \nu(K \cap E) + \underbrace{\nu(K \cap E^c)}_{\leq \nu(E^c)=0} = \int_{K \cap E} f \, \mathrm{d}\mu = \int_K f \, \mathrm{d}\mu$$

as ν was assumed to be positive and $\mu(E^c) = 0$ so we have $d\nu = f d\mu$ in general so we are done. In the other case where $\mu(E^c) > 0$ and $\nu(E^c) = \infty$ we can set f to be ∞ on E^c, then we see that $\nu(E \cap K) = \int_{E \cap K} f d\mu$ as usual and either $\nu(E^c \cap K) = \mu(E^c \cap K) = 0$ or $\nu(E^c \cap K) = \infty$ and $\mu(E^c \cap K) > 0$ so we see that $\int_{E^c \cap K} f d\mu$ and $\nu(E^c \cap K)$ are either both infinite or both 0 by part (c) of the hint as $E^c \cap K$ is disjoint from E, hence we have the equality,

$$\nu(K) = \nu(E \cap K) + \nu(E^c \cap K) = \int_{E \cap K} f d\mu + \int_{E^c \cap K} f d\mu = \int_K f d\mu$$

so we are done. □

Problem 3.15. A measure μ on $(X, \mathcal{M})$ is called decomposable if there is a family $\mathfrak{F} \subset \mathcal{M}$ with the following properties: (i) $\mu(F) < \infty$ for all $F \in \mathfrak{F}$; (ii) the members of $\mathfrak{F}$ are disjoint and their union is X; (iii) if $\mu(E) < \infty$ then $\mu(E) = \sum_{F \in \mathfrak{F}} \mu(E \cap F)$; (iv) if $E \subset X$ and $E \cap F \in \mathcal{M}$ for all $F \in \mathfrak{F}$ then $E \in \mathcal{M}$.

a. Every σ-finite measure is decomposable.

b. If μ is decomposable and ν is any signed measure on $(X, \mathcal{M})$ such that $\nu \ll \mu$, there exists a measurable $f : X \to [-\infty, \infty]$ such that $\nu(E) = \int_E f \, d\mu$ for any E that is σ-finite for μ, and $|f| < \infty$ on any $F \in \mathfrak{F}$ that is σ-finite for ν. (Use Exercise 14 if ν is not σ-finite.)

Solution.

a. It can be easily shown that $\mathfrak{F} = \{E_n : n \in \mathbb{N}\}$ works where E_n partition X while being μ-finite.

b. Let $\mathfrak{F}$ be as described in the problem. As $\mu(F) < \infty$ for any $F \in \mathfrak{F}$ we can find a measurable f_F on F such that $\nu(E \cap F) = \int_{E\cap F} f d\mu$ for all $E \in \mathcal{M}$ by the previous problem and if F is σ-finite for ν we can clearly force $|f_F| < \infty$ (in the proof presente in the book we can take all f_j's which are zero outside A_j to be real valued everywhere as shown in case I which makes their sum real valued too as the A_j are disjoint.) as in the proof of the radon-nikodym theorem. We claim that $f := f_F$ on $F \in \mathfrak{F}$ is measurable and it has the properties needed. For any $E \in \mathcal{B}_{\overline{\mathbb{R}}}$ we see that $f^{-1}(E) = \bigcup_{F\in\mathfrak{F}} f^{-1}(E) \cap F$ and each $f^{-1}(E) \cap F = f|_F^{-1}(E) \cap F = f_F^{-1}(E) \cap F \in \mathcal{M}$ as f_F is measurable which implies that $f^{-1}(E)$ by property (iv). Now, for any E that is σ-finite for μ, suppose $E = \cup E_n$ (disjoint union) with $\mu(E_n) < \infty$, then we see that $\mu(E_n) = \sum_{\mathfrak{F}} \mu(E_n \cap F) < \infty$ and it is well known that only atmost countably many $\mu(E_n \cap F)$ are nonzero, so we can find $\{F_m\} \subseteq \mathfrak{F}$ such that $\mu(E_n) = \sum_{m=1}^{\infty} \mu(E_n \cap F_m)$ and thus we also see that $\mu(E_n \setminus (\cup_m(E_n \cap F_m))) = 0$ which, by absolute continuity implies $\nu(E_n) = \nu(\cup_m(F_m \cap E_n) = \sum_{m=1}^{\infty} \mu(E_n \cap F_m) = \sum_{m=1}^{\infty} \int_{E_n\cap F_m} f d\mu = \int_{E_n} f d\mu$ and thus, $\nu(E) = \sum_1^{\infty} \nu(E_n) = \sum_1^{\infty} \int_{E_n} f d\mu = \int_E f d\mu$.

□

Problem 3.16. Suppose that μ, ν are measures on $(X, \mathcal{M})$ with $\nu \ll \mu$, and let $\lambda = \mu + \nu$. If $f = d\nu/d\lambda$, then $0 \le f < 1$ λ-a.e. and $d\nu/d\mu = f/(1-f)$.

Solution. (note: we treat the derivatives as equivalence classes here.) Firstly, it is obvious that $f \ge 0$ λ-a.e We have $\int_E d\nu = \nu(E) = \int_E f d(\mu+\nu) \le \int_E d\nu + \int_E d\mu = \int d(\mu+\nu)$, it can then be easily shown that $f \le 1$ λ-a.e. If we had $f = 1$ over E with $\lambda(E) = \nu(E) + \mu(E) > 0$ then we see that, $\nu(E) = \int_E f d\lambda = \lambda(E)$ which implies that $\mu(E) = 0$ but then as $\nu \ll \mu$ we have $\nu(E) = 0$ too contradicting non-nullity, this proves that $f < 1$ λ-a.e. too so we have proven the first claim. We clearly have $\nu \ll \mu \ll \lambda$ and $\lambda \ll \mu$. Thus, $g = d\mu/d\lambda$ exists and by Proposition 3.9 we see that $f = d\nu/d\lambda = (d\nu/d\mu)(d\mu/d\lambda) = g(d\nu/d\mu)$ and substituting these in we get $d\nu/d\mu = f/g = f/(1-f)$ as $f + g = d\nu/d\lambda + d\mu/d\lambda = d(\mu+\nu)/d\lambda = d\lambda/d\lambda = 1$. □

Problem 3.17. Let $(X, \mathcal{M}, \mu)$ be a σ-finite measure space, $\mathcal{N}$ a sub-σ-algebra of $\mathcal{M}$, and $\nu = \mu|_{\mathcal{N}}$ is σ-finite (This condition is missing in the book statement, without which this is false: Consider the lebesgue measure on $\mathbb{R}$ as the measure space and take $\mathcal{N} = \{\emptyset, \mathbb{R}\}, f = \chi_{[0,1]}$.) If $f \in L^1(\mu)$, there exists $g \in L^1(\nu)$ (thus g is $\mathcal{N}$-measurable) such that $\int_E f\ d\mu = \int_E g\ d\nu$ for all $E \in \mathcal{N}$; if g' is another such function then $g = g'$ ν-a.e. (In probability theory, g is called the conditional expectation of f on $\mathcal{N}$.)

Solution. As $f \in L^1$ we have that $f d\mu$ is a signed measure and it is clearly still a measure restricted to $\mathcal{N}$ where we see that $(f d\mu)|_{\mathcal{N}} \ll \nu = \mu|_{\mathcal{N}}$ and as $\mathcal{N}$ is sigma finite for ν we can apply Problem 3.14 to deduce existence of some g such that $(f d\mu)|_{\mathcal{N}} = g d\nu$ which implies $\int_E f d\mu = \int_E g d\nu$ for all $E \in \mathcal{N}$. □

Problem 3.18. Prove Proposition 3.13c.

Solution. Let g be as in the definition ($d\nu = g d\mu$ with μ being positive) of $|\nu|$ then we see that $|f| d|\nu| = |f||g| d\mu$ and we also have $|f| d\nu = |f| g d\mu$ and we have that,

$$\left|\int f \,\mathrm{d}\nu\right| = \left|\int fg \,\mathrm{d}\mu\right| \le \int |fg| d\mu = \int |f| d|\nu|$$

$\square$

Problem 3.19. If ν, λ are complex measures and μ is a positive measure, then $\nu \perp \mu$ iff $|\nu| \perp |\mu|$, and $\nu \ll \lambda$ iff $|\nu| \ll \lambda$.

Solution. If $\nu \perp \mu$ then by definition $\nu_a \perp \mu_b$ where $a, b = i, r$ following the notation in the book. By Problem 3.2 this implies that $|\nu_a| \perp |\mu_b|$ where $a, b = i, r$ and using Problem 3.9 twice we see that $|\nu_i| + |\nu_r| \perp |\mu_r| + |\mu_i|$. We see that $\nu \ll |\nu_i| + |\nu_r|$ as $\nu_a \ll |\nu_i| + |\nu_r|$ where $a = i, r$ and similarly for μ which implies that $\nu \perp |\mu_i| + |\mu_r|$ and by a similar arguement, $\nu \perp \mu$. Now if $|\nu| \perp |\mu|$ then, as $\nu \ll |\nu|$ and $\mu \ll |\mu|$ we have that $\nu \perp \mu$. $\square$

Problem 3.20. If ν is a complex measure on $(X, \mathcal{M})$ and $\nu(X) = |\nu|(X)$, then $\nu = |\nu|$.

Solution. We see that $\nu(X) = |\nu(X)| = |\nu(E) + \nu(X \setminus E)| \leq |\nu(E)| + |\nu(X \setminus E)| \leq |\nu|(E) + |\nu|(X \setminus E) = |\nu|(X) = \nu(X)$ so all the inequalities must be equalities, which implies that $|\nu(E)| = |\nu|(E)$ as $|\nu(E)| \leq |\nu|(E)$ for all $E \in \mathcal{M}$. As we have $|\nu(E)| + |\nu(X \setminus E)| = |\nu(E) + \nu(X \setminus E)|$ which is real, those two complex numbers share the same arguement (unless 0 but that case is trivial) and as $\nu(E) + \nu(X \setminus E) = \nu(X)$ is real too, they must have no complex part making them real which implies that, $\nu(E) = |\nu(E)| = |\nu|(E)$ proving the claim. $\square$

Problem 3.21. Let ν be a complex measure on $(X, \mathcal{M})$. If $E \in \mathcal{M}$ define

$$\mu_1(E) = \sup\left\{\sum_1^n |\nu(E_j)| : n \in \mathbb{N}, E_1, ..., E_n \text{ disjoint}, E = \bigcup_1^n E_j\right\},$$

$$\mu_2(E) = \sup\left\{\sum_1^\infty |\nu(E_j)| : E_1, E_2, ... \text{ disjoint}, E = \bigcup_1^\infty E_j\right\},$$

$$\mu_3(E) = \sup\left\{\left|\int_E f \, d\nu\right| : |f| \leq 1\right\}.$$

Then $\mu_1 = \mu_2 = \mu_3 = |\nu|$. (First show that $\mu_1 \leq \mu_2 \leq \mu_3 \cdot$ To see that $\mu_3 = |\nu|$, let $f = \overline{d\nu/d|v|}$ and apply Proposition 3.13. To see that $\nu_3 \leq \nu_1$, approximate f by simple functions.)

Solution. $\mu_1 \leq \mu_2$ is obvious and $\mu_2 \leq \mu_3$ holds as for any given E and any partition $\{E_n\}$ of E we can define

$$f = \sum_n \frac{\overline{\nu(E_n)}}{|\nu(E_n)|} \cdot \chi_{E_n}$$

where the sum is over all n for which $\nu(E_n) \neq 0$, then we clearly see that $|f| \leq 1$ and $\int_E f \, d\nu = \sum_1^\infty \int_{E_n} f \, d\nu = \sum_1^\infty |\nu(E_n)|$ as $|z| = \overline{z}z/|z|$ for nonzero z. Clearly $\mu_3(E) \leq \int_E d|\nu| = |\nu|(E)$ by Proposition 3.13c and with f as in the hint (this can be made to have absolute value atmost 1 everywhere after changing it on a null set, not affecting any of the integrals below) we see that,

$$\int_E \overline{\frac{d\nu}{d|\nu|}} d\nu = \int_E \overline{\frac{d\nu}{d|\nu|}} \frac{d\nu}{d|\nu|} d|\nu| = \int_E |f|^2 d||\nu| = \int_E d|\nu| = |\nu|(E)$$

as $|f| = 1$ $|\nu|$-a.e. so we must have $\mu_3 = |\nu|$. Now, we can find a sequence of simple functions ϕ_n converging f with $|\phi_n| \leq |f| \leq 1$ and convergence can be made uniform here as f is bounded on E, this clearly implies that $\lim |\int_E \phi_n \, d\nu| = |\int_E f \, d\nu|$ and thus, as $|\int_E \phi_n \, d\nu| \leq \mu_1(E)$ due to linearity and the triangle inequality we must have $\lim_n |\int_E f \, d\nu| \leq \mu_1(E)$ too so we are done. □

Problem 3.22. If $f \in L^1(\mathbb{R}^n)$, $f \neq 0$, there exist $C, R > 0$ such that $Hf(x) \geq C|x|^{-n}$ for $|x| > R$. Hence $m(\{x : Hf(x) > \alpha\}) \geq C'/\alpha$ when α is small, so the estimate in the maximal theorem is essentially sharp.

Solution.

□

Problem 3.23. A useful variant of the Hardy-Littlewood maximal function is $H^*f(x) = \sup\{\frac{1}{m(B)} \int_B |f(y)|dy : B \text{ is a ball and } x \in B\}$. Show that $Hf \leq H^*f \leq 2^n Hf$.

Solution.

□

Problem 3.24. If $f \in L^1_{loc}$ and f is continuous at x, then x is in the Lebesgue set of f.

Solution.

□

Problem 3.25. If E is a Borel set in $\mathbb{R}^n$, the density $D_E(x)$ of E at x is defined as $D_E(x) = \lim_{r \to 0} \frac{m(E \cap B(r,x))}{m(B(r,x))}$ whenever the limit exists.

a. Show that $D_E(x) = 1$ for a.e. $x \in E$ and $D_E(x) = 0$ for a.e. $x \in E^c$.

b. Find examples of E and x such that $D_E(x)$ is a given number $\alpha \in (0, 1)$, or such that $D_E(x)$ does not exist.

Solution.

□

Problem 3.26. If λ and μ are positive, mutually singular Borel measures on $\mathbb{R}^n$ and $\lambda + \mu$ is regular, then so are λ and μ.

Solution.

□

Problem 3.27. Verify the assertions in Examples 3.25.

Solution.

□

Problem 3.28. If $F \in NBV$, let $G(x) = |\mu_F|((-\infty, x])$. Prove that $|\mu_F| = \mu_{T_F}$ by showing that $G = T_F$ via the following steps.

a. From the definition of T_F, $T_F \leq G$.

b. $|\mu_F(E)| \leq \mu_{T_F(E)}$ when E is an interval, and hence when E is a Borel set.

c. $|\mu_F| \leq \mu_{T_F}$, and hence $G \leq T_F$.

Solution.

□

Problem 3.29. If $F \in NBV$ is real-valued, then $\mu_F^+ = \mu_P$ and $\mu_F^- = \mu_N$ where P and N are the positive and negative variations of F.

Solution.

□

Problem 3.30. Construct an increasing function on $\mathbb{R}$ whose set of discontinuities is $\mathbb{Q}$.

Solution.

□

Problem 3.31. Let $F(x) = x^2 \sin(x^{-1})$ and $G(x) = x^2 \sin(x^{-2})$ for $x \neq 0$, and $F(0) = G(0) = 0$.

a. F and G are differentiable everywhere (including $x = 0$).

b. $F \in BV([-1, 1])$, but $G \notin BV([-1, 1])$.

Solution.

□

Problem 3.32. If $F_1, F_2, ..., F \in NBV$ and $F_j \to F$ pointwise, then $T_F \leq \liminf T_{F_j}$.

Solution.

□

Problem 3.33. If F is increasing on $\mathbb{R}$, then $F(b) - F(a) \geq \int_a^b F'(t)dt$.

Solution.

□

Problem 3.34. Suppose $F, G \in NBV$ and $-\infty < a < b < \infty$.

a. By adapting the proof of Theorem 3.36, show that

$$\int_{[a,b]} \frac{F(x) + F(x-)}{2} dG(x) + \int_{[a,b]} \frac{G(x) + G(x-)}{2} dF(x) = F(b)G(b) - F(a-)G(a-)$$

b. If there are no points in $[a, b]$ where F and G are both discontinuous, then

$$\int_{[a,b]} F \, dG + \int_{[a,b]} G \, dF = F(b)G(b) - F(a-)G(a-).$$

Solution.

□

Problem 3.35. If F and G are absolutely continuous on $[a, b]$, then so is FG, and

$$\int_a^b (FG' + GF')(x)dx = F(b)G(b) - F(a)G(a).$$

Solution.

□

Problem 3.36. Let G be a continuous increasing function on $[a, b]$ and let $G(a) = c$, $G(b) = d$.

a. If $E \subset [c, d]$ is a Borel set, then $m(E) = \mu_G(G^{-1}(E))$.

b. If f is a Borel measurable and integrable function on $[c, d]$, then $\int_c^d f(y)dy = \int_a^b f(G(x))dG(x)$. In particular, $\int_c^d f(y)dy = \int_a^b f(G(x))G'(x)dx$ if G is absolutely continuous.

c. The validity of (b) may fail if G is merely right continuous rather than continuous.

Solution.

□

Problem 3.37. Suppose $F : \mathbb{R} \to \mathbb{C}$. There is a constant M such that $|F(x) - F(y)| \leq M|x-y|$ for all $x, y \in \mathbb{R}$ (that is, F is Lipschitz continuous) iff F is absolutely continuous and $|F'| \leq M$ a.e.

Solution.

□

Problem 3.38. If $f : [a, b] \to \mathbb{R}$, consider the graph of f as a subset of $\mathbb{C}$, namely, $\{t + if(t) : t \in [a, b]\}$. The length L of this graph is by definition the supremum of the lengths of all inscribed polygons.

a. Let $F(t) = t + if(t)$; then L is the total variation of F on $[a, b]$.

b. If f is absolutely continuous, $L = \int_a^b [1 + f'(t)^2]^{1/2} dt$.

Solution.

□

Problem 3.39. If $\{F_j\}$ is a sequence of nonnegative increasing functions on $[a, b]$ such that $F(x) = \sum_1^\infty F_j(x) < \infty$ for all $x \in [a, b]$, then $F'(x) = \sum_1^\infty F_j'(x)$ for a.e. $x \in [a, b]$.

Solution.

□

Problem 3.40. Let F denote the Cantor function on $[0, 1]$ (see §1.5), and set $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x > 1$. Let $\{[a_n, b_n]\}$ be an enumeration of the closed subintervals of $[0, 1]$ with rational endpoints, and let $F_n(x) = F((x - a_n)/(b_n - a_n))$. Then $G = \sum_1^\infty 2^{-n} F_n$ is continuous and strictly increasing on $[0, 1]$, and $G' = 0$ a.e.

Solution.

□

Problem 3.41. Let $A \subset [0,1]$ be a Borel set such that $0 < m(A \cap I) < m(I)$ for every subinterval I of $[0,1]$.

a. Let $F(x) = m([0,x] \cap A)$. Then F is absolutely continuous and strictly increasing on $[0,1]$, but $F' = 0$ on a set of positive measure.

b. Let $G(x) = m([0,x] \cap A) - m([0,x] \setminus A)$. Then G is absolutely continuous on $[0,1]$, but G is not monotone on any subinterval of $[0,1]$.

Solution.

□

Problem 3.42. A function $F : (a,b) \to \mathbb{R}(-\infty \leq a < b \leq \infty)$ is called convex if $F(\lambda s + (1-\lambda)t) \leq \lambda F(s) + (1-\lambda)F(t)$ for all $s,t \in (a,b)$ and $\lambda \in (0,1)$.

a. F is convex iff for all $s,t,s',t' \in (a,b)$ such that $s \leq s' < t'$ and $s < t \leq t'$,

$$\frac{F(t) - F(s)}{t - s} \leq \frac{F(t') - F(s')}{t' - s'}$$

b. F is convex iff F is absolutely continuous on every compact subinterval of (a,b) and F' is increasing (on the set where it is defined).

c. If F is convex and $t_0 \in (a,b)$, there exists $\beta \in \mathbb{R}$ such that $F(t) - F(t_0) \geq \beta(t - t_0)$ for all $t \in (a,b)$.

d. (Jensen's Inequality) If $(X, \mathcal{M}, \mu)$ is a measure space with $\mu(X) = 1$, $g : X \to (a,b)$ is in $L^1(\mu)$, and F is convex on (a,b), then

$$F\left(\int g \, d\mu\right) \leq \int F \circ g \, d\mu.$$

Solution.

□